

METHOD

How It Works — From Returns to a Tested Risk Number

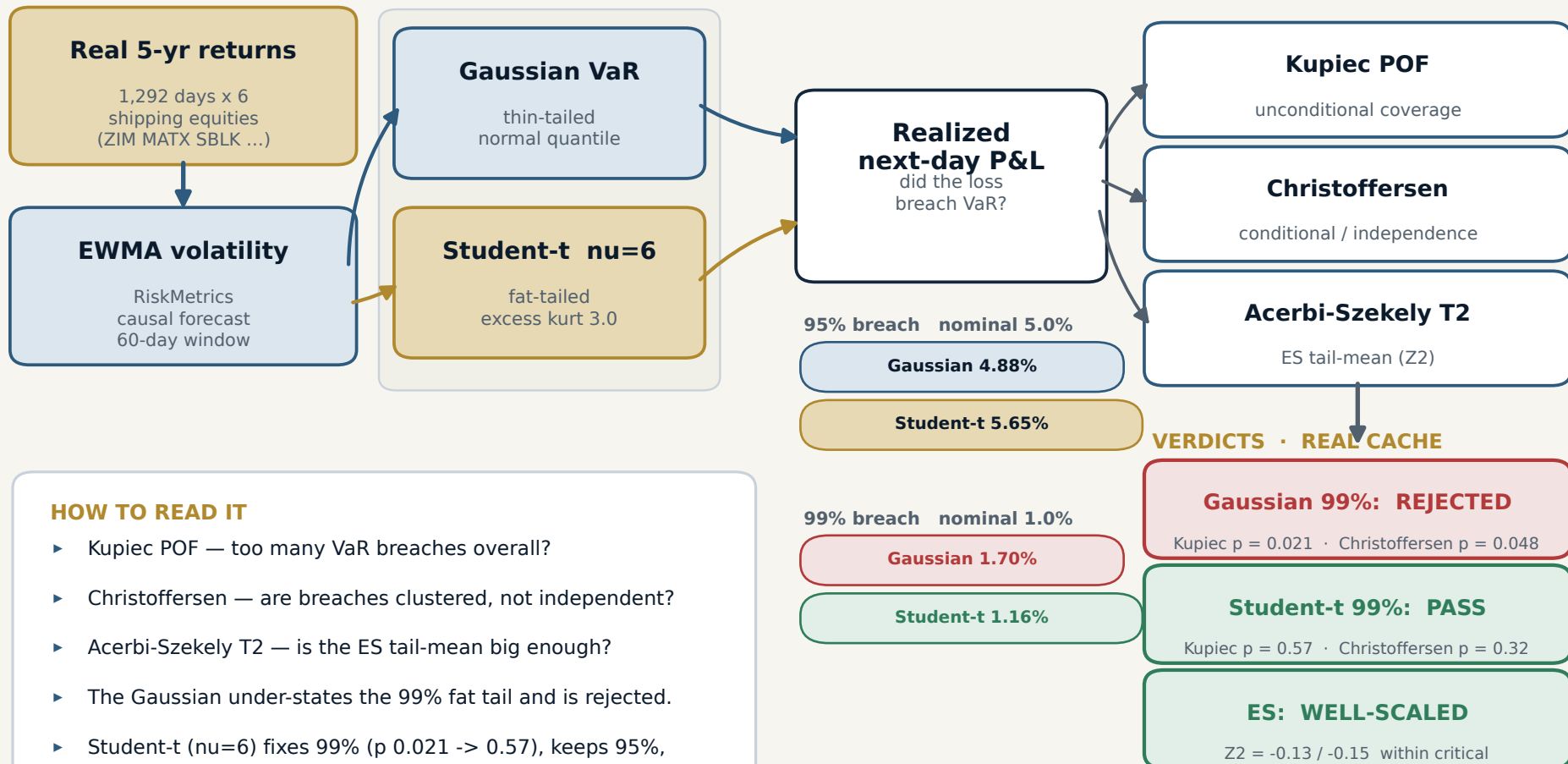
Real cache: 1,292 trading days x 6 shipping equities, 5y, 60-day EWMA window. VaR / ES coverage-tested against realized P&L.

1 · INPUT

2 · TAIL MODELS

3 · REALIZED P&L

4 · VALIDATION GATES



HOW TO READ IT

- ▶ Kupiec POF — too many VaR breaches overall?
- ▶ Christoffersen — are breaches clustered, not independent?
- ▶ Acerbi-Szekely T2 — is the ES tail-mean big enough?
- ▶ The Gaussian under-states the 99% fat tail and is rejected.
- ▶ Student-t (nu=6) fixes 99% (p 0.021 -> 0.57), keeps 95%,
- ▶ and the ES reads honest (Z2 ~ 0). Tested, not asserted.

Coverage tests run nightly · risk_lab — the 2 real-data validators